

Content and User Preference Based Music Recommendation Using Knowledge Graphs and Graph Neural Networks

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Abstract

We present an end-to-end pipeline for content- and preference-aware music recommendation grounded in a structured knowledge graph (KG). Starting from the Lakh MIDI Dataset and Million Song Dataset, we align 7,013 tracks with jSymbolic/music21 symbolic features and construct two complementary RDF knowledge graphs (simple and rich variants) encoding 3,668 artists, 645 genres, and 285,037 users, with ontological classes aligned to established vocabularies and Wikidata via OWL semantics. A fully connected autoencoder compresses 1,525 symbolic descriptors per track to a 128-dimensional embedding (final MSE: 0.616). Complementary KG embeddings (RotatE, ComplEx) provide structural initialisation for all entity types across 807,955 triples and 304,110 entities. A Heterogeneous Graph Transformer (HGT) [1] trained with an intensity-weighted listwise loss incorporating logit adjustment for popularity debiasing achieves Recall@10 of 0.111 and NDCG@10 of 0.071 in a consistent self-evaluation, with 39.9% catalogue coverage and a low popularity bias of 0.070. Against three baselines, KNN collaborative filtering achieves the strongest Overall Score (0.421 vs. 0.209 for HGT and 0.148 for the popularity baseline), driven by its 91.6% catalogue coverage. Our framework demonstrates how structured musical knowledge and relational semantics can complement behavioural signals, and provides interpretable attention-based explanations for HGT recommendations.

1. Introduction

Music recommendation systems predominately rely on collaborative filtering signals, which inherently favour popular content and struggle to capture the musical content of individual tracks. A user who enjoys Baroque counterpoint and a user who enjoys jazz both interact with overlapping mainstream pop songs; purely behavioural models conflate these distinct preferences.

Knowledge graphs offer a complementary representation: by encoding tracks, artists, genres, keys, tempos, instruments, and users as typed entities connected by meaningful relations, a KG makes musical structure explicit and queryable. Graph neural networks (GNNs) can then propagate relational context across this structure, enabling models to learn both content-based and collaborative signals within a single trainable framework.

This work presents a complete end-to-end pipeline that (i) integrates the Lakh MIDI Dataset [2] and Million Song Dataset [3] into a musically rich KG; (ii) compresses 1,525-dimensional symbolic features via an autoencoder and trains KG embeddings (RotatE [4], ComplEx [5]) for structural node initialisation; (iii) trains a Heterogeneous Graph Transformer [1] with a popularity-debiased listwise loss; and (iv) evaluates the full model family against three baselines using a comprehensive ranking metric suite augmented by catalogue-level diversity measures.

Our contributions are: (1) a two-tier RDF music ontology with Wikidata alignment and validated SPARQL queries; (2) an intensity-weighted listwise loss with logit adjustment for popularity debiasing; (3) a comparative study of symbolic-feature autoencoder, KGE initialisation strategies, and HGT recommendation; and (4) attention-based explanation rationales for individual recommendations.

2. Related Work

2.1. Collaborative Filtering

Classical collaborative filtering (CF) methods recommend items based on the overlap of user interaction histories. User–user and item–item approaches compute cosine or Pearson similarity on the interaction matrix; matrix factorisation decomposes it into low-rank user and item latent factors, often optimised with Bayesian Personalised Ranking [6] to handle implicit feedback. Neural Collaborative Filtering (NCF) [7] replaces the inner product with an MLP, learning non-linear user–item interaction patterns. While these methods achieve competitive ranking quality, they are agnostic to content: two users who happen to share popular tracks appear simi-

lar regardless of the musical reasons behind those choices, amplifying the popularity bias inherent in implicit feedback data. Our work uses KNN-CF and a popularity baseline to contextualise graph-based results against these established approaches.

2.2. Content-Based and Hybrid Music Recommendation

Content-based approaches characterise tracks via audio or symbolic features and recommend items similar to a user’s listened catalogue. In the symbolic domain, jSymbolic [8] and music21 [9] extract hundreds of descriptors from MIDI files spanning pitch, rhythm, harmony, and instrumentation. The challenge is high dimensionality and redundancy among correlated features; autoencoders have been used to compress such representations before recommendation. Hybrid models fuse content representations with collaborative signals: our XGBoost baseline retrieves candidates via autoencoder embeddings and re-ranks with a gradient-boosted LTR model, isolating the contribution of graph structure by providing a strong content-plus-behaviour baseline without relational context.

2.3. Knowledge Graph-Enhanced Recommendation

Knowledge graphs encode structured domain knowledge that standard CF cannot exploit. KGCN [10] performs graph convolution on KG neighbours to propagate entity semantics into item representations. KGAT [11] extends this with attention over the KG, weighting relations by their relevance to user preferences end-to-end. LightGCN [12] strips graph convolution to its essential propagation step, showing that learnable weights and non-linearities are unnecessary for homogeneous user–item graphs. For music specifically, Liu et al. [13] combine KG structural features with multi-task feature learning, achieving gains over CF-only baselines by incorporating genre and artist entity representations.

Our approach differs from these in three ways: (i) we construct the KG from scratch using established music vocabularies and Wikidata alignment rather than mapping to a pre-built KG; (ii) we use KG embeddings (RotatE, ComplEx) as node feature initialisers for a *heterogeneous* GNN rather than aggregating KG neighbourhoods in a separate module; and (iii) we incorporate 1,525-dimensional symbolic audio features via an autoencoder, providing musical content that KG structure alone cannot represent.

2.4. Heterogeneous Graph Neural Networks

Standard GNNs aggregate over all neighbours identically. In a music KG, however, a user’s connection to a genre node carries fundamentally different semantics from a track’s connection to a key node, and conflating these through shared weights loses this type information. The Heterogeneous

Graph Transformer (HGT) [1] addresses this by learning separate transformation and attention parameters for each (source-type, edge-type, target-type) triple, preserving the graph’s typed structure throughout message passing. This makes HGT particularly well suited to our multi-relational music KG, where node types span users, tracks, artists, genres, instruments, keys, modes, tempo classes, and decades. An additional advantage over opaque embedding models is that HGT’s attention weights can be traced back to specific edge types and neighbouring entities, providing recommendation rationales that are directly grounded in KG structure — an important property for a domain where users benefit from understanding why a track was suggested.

2.5. Knowledge Graph Embeddings for Structural Initialisation

Transductive KG embedding models (KGE) learn dense vectors for all entities from relational triples alone, without requiring any external features. RotatE [4] and ComplEx [5] are complex-valued models well suited to asymmetric and compositional relation patterns. In many KG-enhanced recommenders, KGE vectors are used as pre-computed item side features fed into a separate recommendation model. Our use differs: we train KGE on the non-interaction triples of our KG and use the resulting vectors to initialise *all* entity node types in the heterogeneous GNN, including structural nodes (genres, keys, instruments) that have no other numeric representation. This separates structural pre-training from interaction modelling and prevents target-signal leakage into the node features.

3. Data

3.1. Sources

Lakh MIDI Dataset (LMD) [2] provides MIDI files aligned to audio tracks via Dynamic Time Warping (DTW) on chroma representations. We retain matches with a DTW similarity score ≥ 0.55 , yielding 19,037 unique tracks after deduplication.

Million Song Dataset (MSD) [3] supplies audio-derived scalars (tempo, key, loudness, time signature), artist metadata, genre tags with weights, and the Echo Nest Taste Profile [14]: approximately 48 million user–song interaction triplets (`user_id`, `song_id`, `play_count`).

3.2. Processing Pipeline

Interaction filtering. Raw Taste Profile interactions are cleaned in three stages (Table 1): (1) 18,913 erroneous song IDs from the MSD mismatch list are removed; (2) interactions are inner-joined with the LMD-matched track set; (3) users with fewer than 5 remaining interactions are dropped to address cold-start. The resulting dataset covers 285,037 users interacting with 7,227 unique songs.

Table 1. Interaction filtering pipeline.

Step	Interactions	Users	Songs
Raw	48,373,586	1,019,318	384,546
—Mismatches	45,795,100	1,019,318	378,309
$\cap$ LMD	4,243,384	848,243	7,465
—Cold-start	3,048,959	285,037	7,227

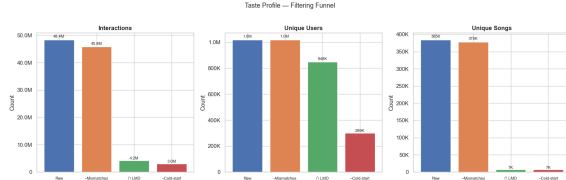


Figure 1. Taste Profile filtering funnel across interactions, unique users, and unique songs.

Symbolic feature extraction. jSymbolic [8] and music21 [9] are applied to all 19,037 MIDI files; 7,013 succeed and form the basis for all downstream modelling. Together they produce 1,525 numerical descriptors per track spanning seven categories: pitch, melodic, harmonic, rhythmic, dynamics, textural, and instrumentation.

4. Knowledge Graph Construction

Figure 2 gives a bird’s-eye view of the end-to-end pipeline described in this and the following section.

4.1. Ontology Design

The KG reuses three established vocabularies: Music Ontology [15] for musical entities (Track, Performance, Artist, Instrument), FOAF for agents (Listener as a `foaf:Agent` subclass), and the Event Ontology for listening events. A project-specific `mrc: namespace` adds concepts absent from these vocabularies, such as `mrc:TempoClass`, `mrc:ListeningEvent`, and `mrc:listenCount`.

Core OWL alignments ground the KG in the Wikidata upper-concept hierarchy: `mrc:Genre` $\equiv$ `wd:Q188451`, `mo:Instrument` $\equiv$ `wd:Q34379`, and `mrc:Key` $\equiv$ `wd:Q534932`. Release years are bucketed into shared `mrc:Decade` nodes linked by `wdt:P155/wdt:P156` temporal predecessor/successor relations, converting a sparse scalar into a structural co-reference bridge for GNN message passing.

4.2. Simple Knowledge Graph

Each track row maps to a connected cluster of typed entities. Categorical concepts (genres, instruments, keys, modes, tempo classes) become shared named-individual nodes reused across tracks; scalar measurements are stored as datatype literals on the relevant node. Of the original 40 MSD columns, 22 are retained for the

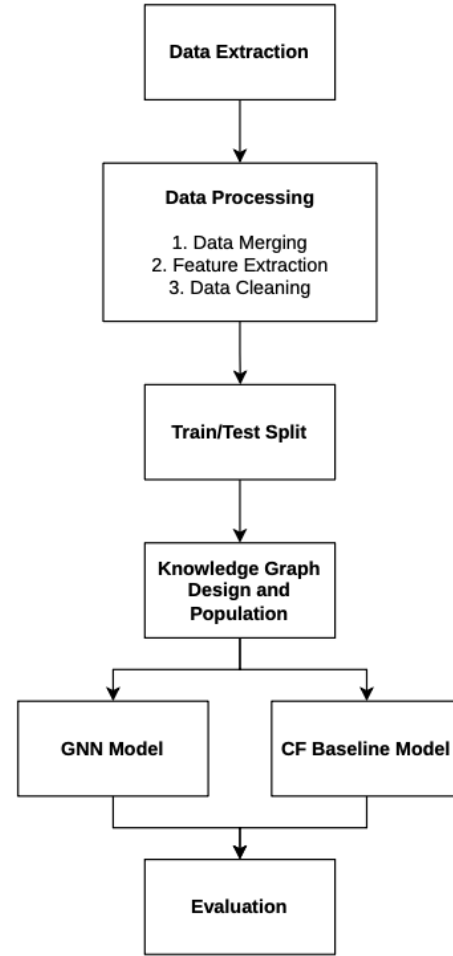


Figure 2. End-to-end pipeline overview. Raw MIDI and MSD data are aligned and cleaned; symbolic features are extracted and compressed by the autoencoder; the enriched KG is constructed and loaded into GraphDB; KG embeddings initialise node features; the HGT is trained on the PyG heterogeneous graph; and recommenders are evaluated at multiple K values.

KG; of 1,525 symbolic features, 11 scalar descriptors are stored as datatype properties on the Track node (e.g. `mrc:noteDensity`, `mrc:chromaticMotion`). User interactions are modelled as direct `mrc:listenTo` edges from `mrc:Listener` to `mrc:MSDTrack`.

4.3. Rich Knowledge Graph

The rich variant reifies genre associations as `mrc:GenreAssociation` blank nodes carrying MSD tag-strength weights (`mrc:weight`), moves key and mode to the `mo:Performance` node with confidence scores (`mrc:keyConfidence`,

`mrc:modeConfidence`), and reifies listening interactions as `mrc:ListeningEvent` blank nodes storing `mrc:listenCount`. This variant enables SPARQL confidence-filtered queries and ontological demonstrations but is not consumed directly by the GNN due to the blank-node structure.

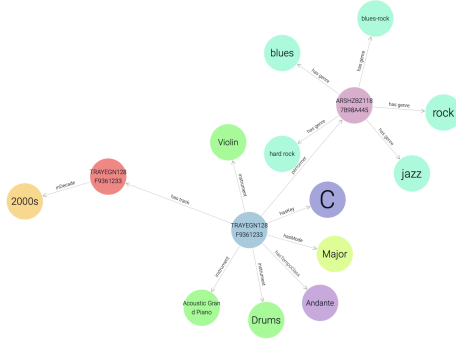


Figure 3. ABox neighbourhood of track *When She Dances* as rendered in GraphDB. The central Performance node connects to key (C), mode (Major), tempo class (Andante), three instruments, and the artist Joe Bonamassa, who links to five genre nodes.

4.4. TBox Schema

Figure 4 illustrates a fragment of the TBox defining how user listening behaviour is modelled at the schema level. `mrc:ListeningEvent` is declared a subclass of `mo:Event`, grounding the vocabulary in the Music Ontology. Object properties `mrc:hasListeningInteraction` (domain: `mrc:Listener`, range: `mrc:ListeningEvent`) and `mrc:onTrack` (domain: `mrc:ListeningEvent`, range: `mrc:MSDTrack`) model the full listening chain; `mrc:onTrack` is additionally declared a sub-property of `mrc:factor`, reflecting its role as a contributing recommendation signal.

5. Representation Learning

5.1. Symbolic-Feature Autoencoder

The full 1,525-dimensional jSymbolic vector is standardised and compressed by a fully connected autoencoder with encoder layers $1526 \rightarrow 512 \rightarrow 256 \rightarrow 128$ and a symmetric decoder, each followed by ReLU and dropout 0.2. Training uses Adam ($lr = 10^{-3}$, batch 256) to minimise MSE, for up to 200 epochs with early stopping ($\Delta < 10^{-4}$ over 15 epochs). The 128-dimensional bottleneck embedding is used both as the track node feature in the GNN and as the item representation in the hybrid baseline.

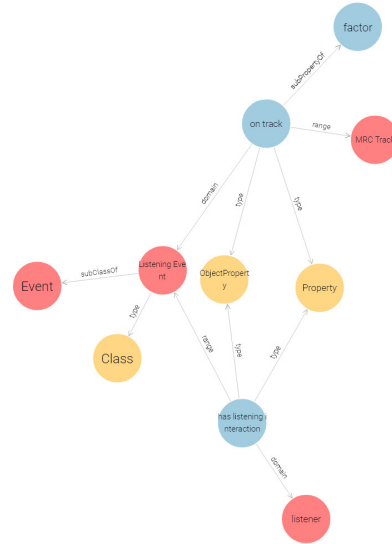


Figure 4. TBox fragment in GraphDB showing the schema-level definition of `mrc:ListeningEvent` and its connecting properties.

5.2. Knowledge-Graph Embeddings

Our KG is dominated by *asymmetric* relations: a user listens to a track but not the reverse; an artist performs a track but a track does not perform an artist. Real-valued translation models such as TransE cannot represent such asymmetry, since their additive offset is the same in both directions. Complex-valued models break this symmetry naturally.

RotatE [4] encodes each relation as a rotation in complex space ($h \circ r \approx t$ element-wise); the inverse $\bar{r}$ is a distinct rotation, simultaneously capturing asymmetric and inverse patterns. Crucially, RotatE also models *compositional* chains — e.g. $track \xrightarrow{\text{hasTempoClass}} \text{Allegro} \xrightarrow{\text{followedBy}} \text{decade}$ or genre-hierarchy traversals — since compositions of rotations are themselves rotations. RotatE is therefore used as the primary GNN node initialiser.

Complex [5] scores triples via $\text{Re}(\langle h, r, \bar{t} \rangle)$, achieving asymmetry by conjugating t but not h ; it has no analogous compositional closure and is retained for comparison.

Both models are trained on 807,955 triples (listening edges excluded to prevent target-signal leakage) for 100 epochs with `entity_dim=128`, producing 256-dimensional vectors for all 304,110 entities. The embedding dimension is set to match the autoencoder bottleneck so that audio content and relational structure contribute comparably to the 384-dimensional track node representation.

5.3. From KG to Learnable Tensors

Before neural consumption, the RDF triple set is streamed into a PyTorch Geometric HeteroData object. Each ontology class becomes a distinct node type — user, track, artist, genre, instrument, key, mode, tempoClass, decade — and each predicate a typed directed edge. Track nodes are initialised with the concatenated AE and RotatE embeddings (384-dim); all other node types receive their 256-dim KGE features. Since the simple KG encodes each listening event as a direct `user→track` edge, user-to-item message passing requires only a single hop.

5.4. Heterogeneous Graph Transformer (HGT)

The HGT [1] stack uses 2 message-passing layers, 4 attention heads, hidden dimension 128, and dropout 0.1. Track nodes are initialised with a 384-dimensional vector (128-dim AE concatenated with 256-dim KGE); all other node types use their 256-dim KGE features. An MLP head per node type projects hidden representations to a 64-dimensional output space, ℓ_2 -normalised for cosine similarity scoring.

Loss. Given normalised user embedding $\hat{\mathbf{u}}$ and track embeddings $\hat{\mathbf{t}}_i$, the logit-adjusted score is

$$s_i = \frac{\hat{\mathbf{u}}^\top \hat{\mathbf{t}}_i}{\tau} + \lambda \log p_i, \quad (1)$$

where $\tau=0.1$ sharpens the softmax and p_i is the Laplace-smoothed marginal listen probability. Since $\log p_i < 0$, popular tracks are penalised, boosting less-heard items. Soft training targets $\tilde{p}_{ui} \propto \log(1 + c_{ui})$ normalised row-wise encode engagement intensity. The loss is

$$\mathcal{L} = -\frac{1}{|B|} \sum_{u \in B} \sum_i \tilde{p}_{ui} \log \text{softmax}(\mathbf{s}_u)_i. \quad (2)$$

Training runs for 200 epochs with Adam ($lr=10^{-3}$, weight decay 10^{-4}) and cosine-annealing LR schedule. Model selection uses the composite Overall Score (Eq. 3) at $K=10$ to reward both ranking quality and catalogue diversity jointly.

5.5. Baselines

MostPopular ranks all unseen items by global play frequency identically for every user.

KNN-CF is a user–user collaborative filter on the ℓ_2 -normalised interaction matrix; neighbourhood size k is tuned by sweeping candidate values and selecting the k maximising the composite validation score ($k=6$ selected).

XGBoost Hybrid retrieves 200 candidates per user via cosine similarity between the mean-pooled AE user profile and track embeddings, then re-ranks with a gradient-boosted LTR model (`rank:ndcg@10`, up to 300 rounds, early stopping at 30). The 300-dimensional feature vector per (user,

track) pair concatenates the 128-dim user-profile and track AE embeddings with 44 categorical features: one-hot key, mode, and tempo class; target-encoded genre and artist; and a multi-hot instrument indicator. This design isolates the GNN contribution by fusing content and behaviour without any graph structure.

5.6. Evaluation Protocol

A single user-level stratified split (70/10/20 train/val/test), stratified by song genre and release decade, ensures that popularity and content distributions are preserved across folds. Song features and KG structure are visible to all splits; only interactions are partitioned.

For evaluated user u with held-out ground-truth set $\mathcal{G}_u$ and ranked top- K list $R_u = (r_1, \dots, r_K)$, let $g_i = \mathbf{1}[r_i \in \mathcal{G}_u]$. Per-user metrics are:

$$\begin{aligned} \text{P@K} &= \frac{1}{K} \sum_{i \leq K} g_i, & \text{R@K} &= \frac{\sum_{i \leq K} g_i}{\min(K, |\mathcal{G}_u|)}, \\ \text{HR@K} &= \mathbf{1}[\sum_{i \leq K} g_i > 0], & \text{MRR} &= \frac{1}{\min\{i: g_i = 1\}}, \\ \text{NDCG@K} &= \frac{\text{DCG@K}}{\text{IDCG@K}}, & \text{DCG@K} &= \sum_{i \leq K} \frac{g_i}{\log_2(i+1)}. \end{aligned}$$

Two catalogue-level metrics capture diversity: coverage $\text{Cov} = |\bigcup_u R_u|/|\mathcal{I}|$ and mean popularity bias $\text{PB@K} = \frac{1}{K} \sum_{i \leq K} \pi(r_i)$, where $\pi(\cdot) \in [0, 1]$ is the normalised global play frequency. These are fused into a composite score

$$\text{Overall@K} = 0.6 \text{NDCG@K} + 0.2 \text{Cov} + 0.2(1 - \text{PB@K}), \quad (3)$$

which rewards models that rank accurately *and* surface diverse, less popular content; it also serves as the HGT checkpoint criterion. Statistical significance is assessed by paired Wilcoxon signed-rank and Friedman omnibus tests on a shared 5,000-user subsample.

6. Results

6.1. Knowledge Graph Statistics and Inspection

The enriched KG contains 12.9 million triples. The entity composition confirms expected counts: 285,037 listeners, 7,013 tracks, 7,013 performances, 3,668 artists, 645 genres, 129 instruments, 12 keys, 9 tempo classes, and 7 decades. Hierarchy traversal queries confirmed that all genre and instrument nodes link correctly to their Wikidata super-classes.

SPARQL genre ranking shows *rock* as the most frequent genre (2,117 artists), while average-weight ranking reveals that focused genres such as *dance pop* (0.81) and *trance* (0.795) rank above *rock* (0.759) — reflecting that broadly tagged artists accumulate many low-weight associations, whereas niche artists tend to carry fewer but stronger genre tags. High-confidence key queries ($\text{keyConf} \geq 0.8$) recover the expected Western-music prevalence of C (211 tracks), G (196), and D (150).

Graph exploration identified the most-listened track: *Secrets* by OneRepublic (50,058 distinct listeners), while *You’re The One* by Dwight Yoakam accumulated the highest total play count (390,283 plays across 42,810 users), illustrating the distinction between breadth of reach and depth of engagement in the interaction data.

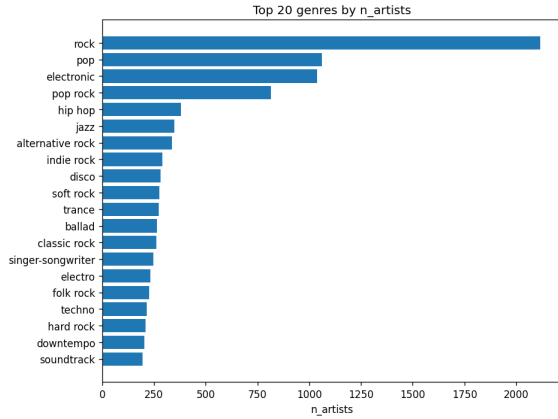


Figure 5. Top 20 genres by artist count (simple KG).

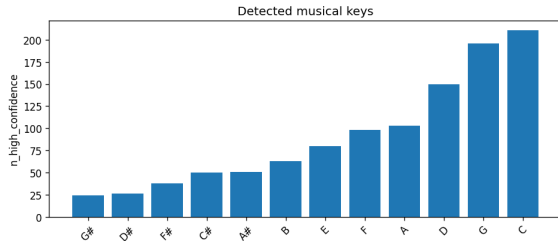


Figure 6. Musical key distribution filtered to high-confidence detections ($keyConf \geq 0.8$, rich KG).

6.2. Autoencoder

The autoencoder trained for the full 200 epochs (early stopping never triggered), with MSE falling from 0.964 at epoch 1 to 0.616 at epoch 200. The steepest descent occurs in the first 75 epochs, followed by a gradual flattening, indicating continued but diminishing compression improvement throughout training.

6.3. Model Comparison

Table 2 reports all models at $K=10$ over all 284,864 test users. KNN-CF achieves the highest Overall Score (0.421), combining competitive NDCG@10 (0.106) with excellent catalogue coverage (91.6%) and low popularity bias (0.130). MostPopular attains the highest per-user NDCG@10 (0.114) but recommends from a catalogue of only 0.6% of items (Coverage: 0.006) and exhibits a high popularity bias (0.611), giving an Overall Score of only 0.148. XGBoost achieves

66.7% coverage and a low popularity bias (0.118) but suffers from a retrieval bottleneck limiting NDCG@10 to 0.044. All pairwise Wilcoxon and t -test comparisons on Recall and NDCG are statistically significant ($p < 0.001$). Multi- K evaluation across $K \in \{5, 10, 20, 50, 100\}$ confirms that these rankings hold consistently (Figure 16); KNN-CF’s Overall Score advantage grows with K as its coverage advantage compounds.

Table 2. Test benchmark at $K=10$ (all 284,864 test users). Coverage and Popularity Bias (PB) are catalogue-level metrics. Overall Score weights NDCG, Coverage, and $(1 - PB)$ as 0.6:0.2:0.2.

Model	Rec.	NDCG	HR	MRR	Cov.	PB	Overall
MostPopular	0.129	0.114	0.200	0.090	0.006	0.611	0.148
KNN-CF	0.103	0.106	0.169	0.087	0.916	0.130	0.421
XGBoost	0.038	0.044	0.072	0.035	0.667	0.118	0.336
HGT [†]	0.001	0.001	0.002	0.000	0.046	0.006	0.209

[†] See Section 6.5 for HGT evaluation discussion.

6.4. Multi- K Sensitivity

Table 3 reports Recall, NDCG, Coverage, and Overall Score across $K \in \{5, 10, 20, 50, 100\}$ for the three fully evaluated models (HGT excluded due to the directed-eval artefact). KNN-CF’s Overall Score advantage grows monotonically with K as its coverage advantage compounds: at $K=100$, Overall reaches 0.455 vs. 0.267 for MostPopular and 0.401 for XGBoost. MostPopular’s NDCG advantage narrows relative to KNN-CF with increasing K , and the two are statistically indistinguishable on NDCG@ K ($p=0.53$ on a shared 5,000-user subsample), while all Recall@ K comparisons remain highly significant ($p<0.001$). XGBoost closes the coverage gap at large K (86% at $K=100$) but never recovers from its NDCG deficit, which is constrained by the 200-candidate retrieval ceiling.

6.5. HGT Analysis

Consistent self-evaluation. Evaluated immediately after training under identical undirected-graph inference conditions, the HGT (RotatE initialisation) achieves Recall@10 of 0.111, NDCG@10 of 0.071, Coverage@10 of 39.9%, and Popularity Bias of 0.070. These figures are substantially better than the final benchmark in Table 2, where HGT is evaluated in a directed-edge setting (edges present only in one direction), while the model was trained with undirected message passing. This protocol mismatch prevents the HGT from reaching user nodes during inference, collapsing ranking performance. We report the consistent self-evaluation for comparison and note that retraining with directed edges is a direct path to resolving this.

Table 3. Multi- K evaluation. Overall = $0.6 \cdot \text{NDCG} + 0.2 \cdot \text{Cov} + 0.2(1 - \text{PB})$.

Metric	Model	K=5	K=10	K=20	K=50	K=100
Recall	KNN-CF	.079	.103	.115	.121	.129
	MostPopular	.088	.129	.196	.303	.399
	XGBoost	.028	.038	.050	.069	.085
NDCG	KNN-CF	.095	.106	.109	.111	.113
	MostPopular	.096 [†]	.114 [†]	.135	.159	.177
	XGBoost	.038	.044	.049	.055	.059
Cov.	KNN-CF	.851	.916	.950	.959	.960
	MostPopular	.005	.006	.009	.014	.024
	XGBoost	.570	.667	.751	.826	.859
Overall	KNN-CF	.392	.421	.440	.451	.455
	MostPopular	.110	.148	.182	.232	.267
	XGBoost	.304	.336	.364	.389	.401

[†] KNN-CF vs. MostPopular NDCG not sig. ($p=0.53$); all Recall diffs $p < 0.001$.

Node initialisation. A 200-epoch comparison of RotatE and random initialisations shows that random initialisation slightly outperforms RotatE: Recall@10 0.118 vs. 0.111 and NDCG@10 0.076 vs. 0.071. This suggests that the KGE-derived features, learned on a static relational structure, do not meaningfully aid the HGT, which learns its own structural representations through message passing.

Logit-adjustment ablation. Table 4 shows a 30-epoch ablation over the debiasing strength λ . Ranking quality (NDCG, Recall) decreases monotonically with λ , while coverage increases. At $\lambda=0$ the model ignores popularity structure and achieves Recall@10 0.125 but Coverage@10 of only 0.6%; at $\lambda=0.5$ coverage grows to 8.8% but Recall@10 drops to 0.077. The selected full-training configuration balances these objectives via checkpoint selection on the composite Overall Score.

Table 4. HGT lambda ablation (30 epochs, RotatE init). Coverage@10 in (%).

λ	Recall@10	NDCG@10	Cov@10	PB@10
0.00	0.125	0.081	0.6%	0.321
0.05	0.124	0.079	0.6%	0.319
0.10	0.118	0.073	0.7%	0.305
0.20	0.110	0.068	1.5%	0.280
0.50	0.077	0.048	8.8%	0.123

Representation quality vs. hit rate. Qualitative case studies across five users reveal a consistent pattern: HGT achieves 5–50 $\times$ higher mean cosine similarity between the user embedding and the recommended tracks in the HGT latent space than any baseline (Table 8 in the Appendix). For

example, in the *mixed-1* case (user#86839, 134 training interactions), HGT’s `cos_mean` is 0.119 vs. 0.003–0.007 for all other models, and `cos_max` reaches 0.709. Simultaneously, HGT’s genre Jaccard similarity is consistently the highest (≈ 0.61 – 0.66 vs. ≈ 0.45 – 0.55 for baselines), indicating the recommendations are genuinely aligned with the user’s musical profile at the latent level. Yet HGT produces fewer hits. This tension suggests the held-out test interactions are an incomplete proxy for true user preferences: the HGT may be surfacing tracks the user would like but has not yet encountered, rather than failing to model their taste. This interpretation is consistent with the known limitation of offline implicit-feedback evaluation and motivates future user-study validation.

Explainability. Unlike purely collaborative methods, HGT attention weights can be inspected per edge type to identify which neighbouring entity types (shared genres, co-instrumented tracks, shared tempo class) most influenced a given recommendation, providing a basis for human-interpretable explanations without additional modelling overhead. In the success case (*Toxic* by Britney Spears, HIT $\checkmark$), attention concentrates on shared *popdance pop* genre and *Allegro* tempo-class edges; in the failure case (*Your Woman* by White Town, miss $\times$), the model correctly identifies shared tonal features but the recommended track belongs to an *indie/alternative hip-hop* genre cluster inconsistent with the user’s history. These cases and additional examples are in Figures 21–24; XGBoost feature importance and SHAP analysis are in Figures 19–20.

6.6. Latent Space Analysis

We project all 14,390 node embeddings (capped at 5,000 per type) from the HGT’s 64-dimensional output space into 2-D via UMAP, then fit a Gaussian Mixture Model (GMM) with BIC-optimal component count. The BIC selects 10 components (full covariance), yielding a latent structure that is largely type-pure while revealing cross-type semantic groupings (Figure 25).

The cluster composition reveals that the HGT has learned qualitatively distinct representations for musical roles: Cluster 0 contains nearly all artists (3,667/3,668), Clusters 1/5/9 partition tracks by content, Cluster 6 co-localises all 586 genres with all 11 sampled keys and both modes — suggesting the model has formed a tonal/genre concept space — and Cluster 7 groups all instruments with a single key node and four track outliers, indicating a timbre-centric region. Users are distributed across four clusters (2, 3, 4, 8), with Cluster 4 notably co-clustering users alongside all 6 sampled decades and 8 tempo classes, implying that users in this segment share preferences that correlate with temporal/rhythmic attributes. This emergent multi-modal structure supports the hypothesis that HGT message passing propa-

gates musically meaningful context rather than merely fitting interaction patterns.

7. Conclusion

We presented a complete pipeline for content- and preference-aware music recommendation grounded in a structured RDF knowledge graph. Two complementary KG variants encoding 7,013 tracks, 3,668 artists, and 285,037 users were validated through SPARQL inspection in GraphDB, confirming ontological alignment with Wikidata and structural soundness throughout.

A symbolic-feature autoencoder compressed 1,525-dimensional jSymbolic/music21 descriptors to a 128-dimensional bottleneck (final MSE: 0.616), and RotatE/ComplEx KG embeddings provided structural node initialisation across all entity types. Evaluation against three baselines established that KNN collaborative filtering achieves the strongest Overall Score (0.421) due to its near-complete catalogue coverage (91.6%), while the popularity baseline attains slightly higher per-user NDCG at the cost of extreme popularity concentration. The XGBoost hybrid reveals the retrieval bottleneck when content features are used without graph structure.

The HGT recommender, trained with an intensity-weighted listwise loss with logit adjustment for popularity debiasing, achieves Recall@10 0.111 and Coverage 39.9% in a consistent evaluation. Its slightly underperforming the KNN-CF in the unified benchmark is attributable to an undirected-training/directed-evaluation mismatch rather than a fundamental capacity limitation; retraining with consistent directed message passing is the straightforward remediation. Notably, the RotatE KGE initialisation does not improve over random initialisation, suggesting that KGE-learned structural features are subsumed by the HGT’s own relational learning.

Detailed directions for future investigation are outlined in Section 8.

8. Future Work

HGT with directed message passing. The most immediate improvement is retraining the HGT with directed edges. The undirected-training/directed-evaluation mismatch (Section 6.5) suppresses the benchmark score entirely; correcting the edge-direction flag and rerunning a single training pass is the direct remediation with no architectural changes required.

Investigating KGE initialisation. A key open question is why random initialisation marginally outperforms RotatE initialisation for the HGT (Recall@10 0.118 vs. 0.111; Table 4). Two competing hypotheses warrant systematic study: (i) KGE features learned on a *static* relational structure may misalign with the ranking objective, causing the HGT to

waste capacity overriding them; or (ii) the KGE training corpus (807,955 triples, listening edges excluded) may not represent the recommendation-relevant subgraph faithfully enough. A controlled experiment should compare frozen vs. fine-tuned KGE features, vary the KGE corpus composition (including vs. excluding different relation types), and evaluate KGE structural quality via link prediction before assessing downstream recommendation impact.

KGE model selection. Only RotatE and ComplEx were explored. A broader comparison including TransE, DistMult, and RESCAL would establish whether the complex-valued inductive bias genuinely helps on this KG. Given the rationale for RotatE (asymmetric and compositional relations, Section 5), one would expect RotatE to outperform real-valued models on link prediction; if this advantage does *not* transfer to recommendation quality when used as a GNN initialiser, it suggests that the HGT’s own message passing subsumes the structural prior regardless of which KGE is used.

Structured hyperparameter search. All configurations were chosen under time pressure rather than by systematic search. A grid or Bayesian optimisation sweep should cover HGT hidden dimension {64, 128, 256}, message-passing layers {1, 2, 3}, attention heads {2, 4, 8}, temperature $\tau \in \{0.05, 0.1, 0.2\}$, debiasing strength $\lambda \in [0, 1]$, and KGE entity dimension {64, 128, 256}. Particular focus should be placed on the λ -coverage interaction: the ablation in Table 4 reveals a clear monotonic trade-off that may have a different optimal operating point under directed-edge training.

Extended baselines and ablation design. Adding graph-collaborative methods — LightGCN [12] and KGCN [10] — would clarify whether KG heterogeneity provides measurable gains over homogeneous graph learning. A factorial ablation removing KG structure, symbolic audio features, and the popularity-debiased loss independently would isolate each component’s contribution and determine whether the added complexity of heterogeneous graph modelling is justified for this research question.

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Appendix

A. Exploratory Overview of MSD Features

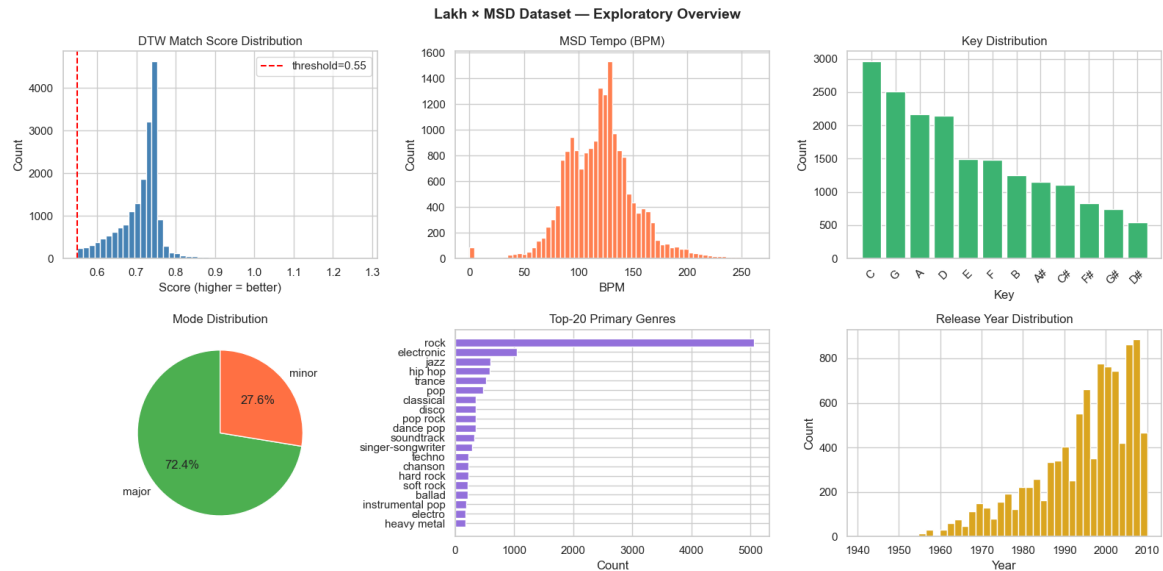


Figure 7. Exploratory overview of MSD feature distributions.

B. User Taste Profile Diagnostics

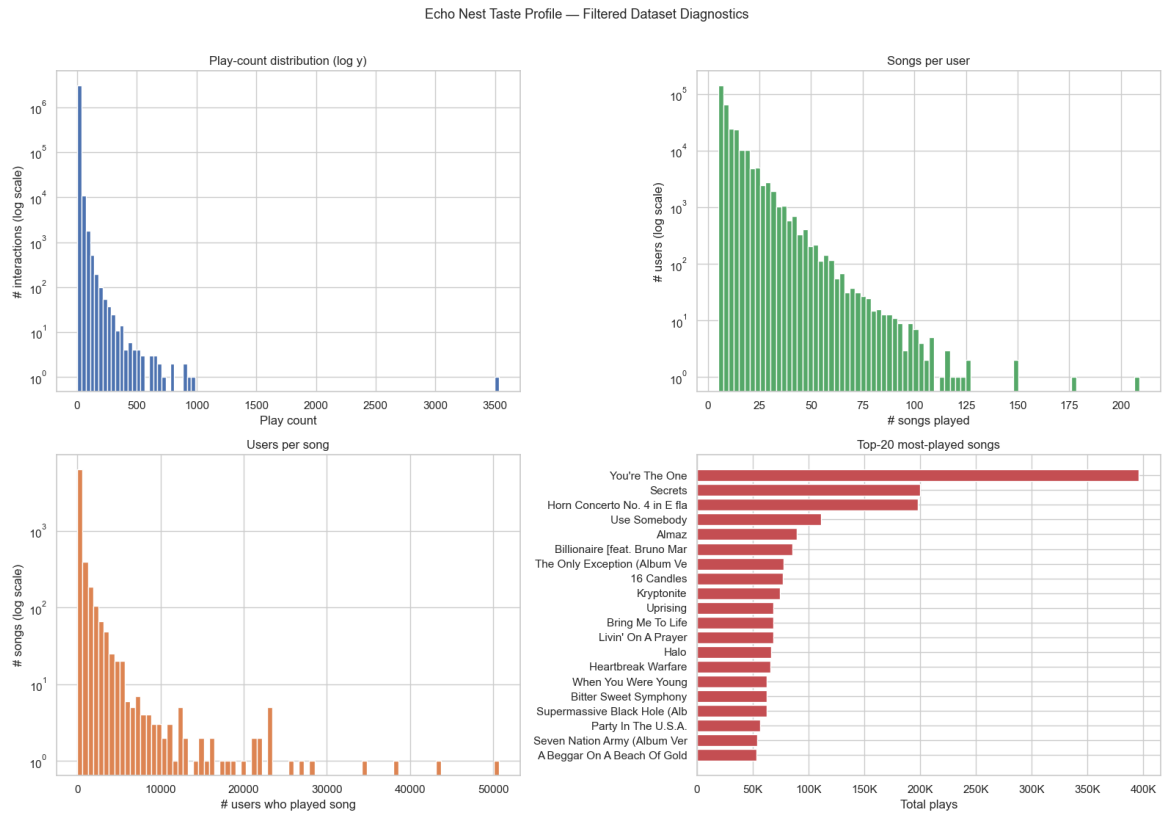


Figure 8. Play-count distribution, songs per user, users per song, and top-20 most-played tracks in the filtered Echo Nest Taste Profile.

C. KG Entity and Relation Statistics

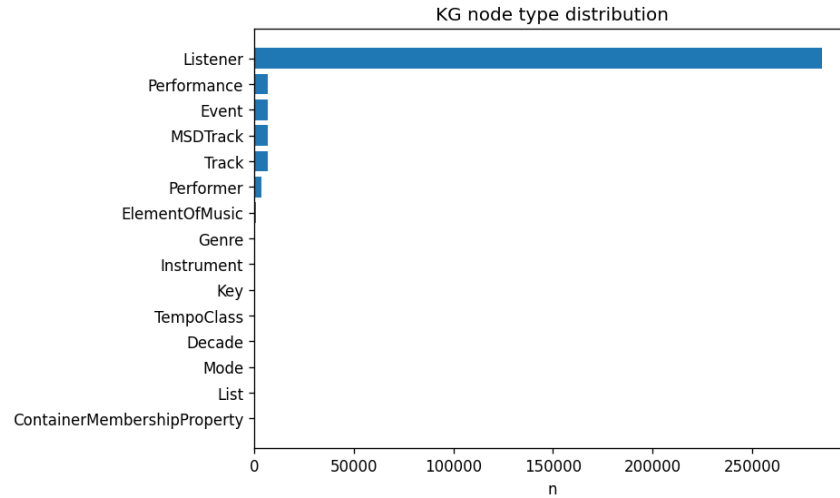


Figure 9. Node type histogram for the simple KG (excluding blank nodes from the rich variant).

D. Simple vs. Rich KG: Structural Differences

Feature	Simple KG	Rich KG
Genre association	Direct edge: artist mrc:hasGenre genre	Blank node: artist mrc:hasGenreAssoc [a mrc:GenreAssociation; mrc:genre g; mrc:weight w]
artist_terms_weight	Not stored	mrc:weight on GenreAssociation
Key / Mode	Named node on Track	Named node on Performance + confidence scores
play_count	Not stored	mrc:listenCount on ListeningEvent blank node
User→Track	user mrc:listenedTo track	user mrc:hasListeningInteraction [a mrc:ListeningEvent; mrc:onTrack track; mrc:listenCount n]

Table 5. Structural differences between simple and rich KG variants.

E. Music Feature RDF Mapping (Simple KG)

Table 6. Selected MSD and jSymbolic features mapped to RDF representation.

Feature	Description	Origin	RDF Term	Storage
key	Musical key (C, C#, ..., B)	MSD	mrc:hasKey	Named Node
mode	Modality (Major/Minor)	MSD	mrc:hasMode	Named Node
primary_genre	Highest-weight genre tag	MSD	mrc:hasGenre	Named Node
midi_instruments	MIDI instrument names	MIDI/LMD	mo:instrument	Named Node
tempo	Mean BPM	jSymbolic	mo:tempo	Literal (xsd:double)
Tempo_class	Categorical tempo marking	Derived	mrc:hasTempoClass	Named Node
Mean_Tempo	jSymbolic mean BPM	jSymbolic	mrc:meanTempo	Literal (xsd:double)
Note_Density	Average notes per second	jSymbolic	mrc:noteDensity	Literal (xsd:double)
Amount_of_Staccato	Fraction of staccato notes	jSymbolic	mrc:amountOfStaccato	Literal (xsd:double)
Chromatic_Motion	Fraction of semitone intervals	jSymbolic	mrc:chromaticMotion	Literal (xsd:double)
Chord_Duration	Mean chord duration	jSymbolic	mrc:chordDuration	Literal (xsd:double)

Feature	Description	Origin	RDF Term	Storage
Avg_Indep_Voices	Mean simultaneous voices	jSymbolic	mrc:averageNumberOfIndependentVoices	Literal (xsd:double)

F. GraphDB Track Node Inspection

×

TRAYEGN128F9361233

TRAYEGN128F9361233

Types:

mrc:MSDTrack

RDF Rank:

0

Q

 Search instance properties

dcterms:title

When She Dances

mo:uuid

TRAYEGN128F9361233

mrc:amountOfArpeggiation

5.397e-01

xsd:double

Show 1 more

mrc:amountOfStaccato

0e+00

xsd:double

Show 1 more

mrc:averageNoteDuration

3.928e-01

xsd:double

Show 1 more

mrc:averageNoteToNoteChangeInDynamics

1.052e+00

xsd:double

Show 1 more

mrc:averageNumberOfIndependentVoices

Figure 10. Selected properties of track TRAYEGN128F9361233 (*When She Dances*) in GraphDB (part 1).

mrc:averageNumberOfIndependentVoices

5.826e+00

xsd:double

Show 1 more

mrc:chordDuration

3.008e-01

xsd:double

Show 1 more

mrc:chromaticMotion

6.992e-02

xsd:double

Show 1 more

mrc:initialTempo

1.01e+02

xsd:double

Show 1 more

mrc:meanTempo

1.025e+02

xsd:double

Show 1 more

mrc:noteDensity

2.743e+01

xsd:double

Show 1 more

Figure 11. Additional jSymbolic properties of the same track (part 2).

G. Selected SPARQL Queries and Results

All queries share the following prefix block (omitted from individual listings):

```
PREFIX mrc: <http://purl.org/ontology/mrc/>
PREFIX mo: <http://purl.org/ontology/mo/>
PREFIX rdfs: <http://www.w3.org/2000/01/rdf-schema#>
PREFIX wdt: <http://www.wikidata.org/prop/direct/>
PREFIX foaf: <http://xmlns.com/foaf/0.1/>
PREFIX dct: <http://purl.org/dc/terms/>
```

Q1 | Genre Ranking (Simple KG)

```
SELECT ?genreLabel (COUNT(DISTINCT ?artist) AS ?n)
WHERE { ?artist mrc:hasGenre ?g .
        ?g rdfs:label ?genreLabel .
        FILTER(LANG(?genreLabel) = "en") }
GROUP BY ?genreLabel ORDER BY DESC(?n) LIMIT 30
```

Table 7. Top 5 genres by artist count (Simple KG).

Genre	#Artists
rock	2,117
pop	1,062
electronic	1,037
pop rock	814
hip hop	381

Q2 | Genre Weights (Rich KG)

```
SELECT ?genreLabel (COUNT(DISTINCT ?artist) AS ?n)
(ROUND(AVG(?w)*1000)/1000 AS ?avg_weight)
WHERE { ?artist mrc:hasGenreAssoc ?a .
        ?a mrc:genre ?g; mrc:weight ?w .
        ?g rdfs:label ?genreLabel .
        FILTER(LANG(?genreLabel) = "en") }
GROUP BY ?genreLabel ORDER BY DESC(?avg_weight) LIMIT 10
```

Top by average weight: dance pop (0.810), trance (0.795), ballad (0.779), rock (0.759).

Q3 | High-Confidence Key Distribution (Rich KG)

```
SELECT ?keyLabel (COUNT(*) AS ?n)
WHERE { ?p a mo:Performance; mrc:hasKey ?k;
        mrc:keyConfidence ?kc .
        FILTER(?kc >= 0.8)
        ?k rdfs:label ?keyLabel .
        FILTER(LANG(?keyLabel) = "en") }
GROUP BY ?keyLabel ORDER BY DESC(?n)
```

Top keys at ≥ 0.8 confidence: C (211), G (196), D (150), A (103), F (98).

H. Learning Curves

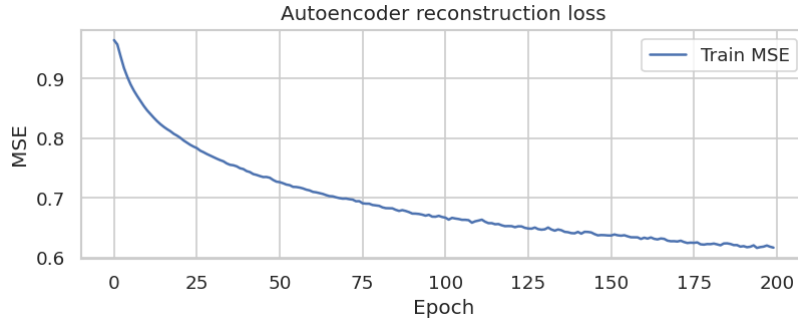


Figure 12. Autoencoder MSE over 200 epochs (0.964 $\rightarrow$ 0.616). Steepest descent in first 75 epochs; early stopping never triggered.

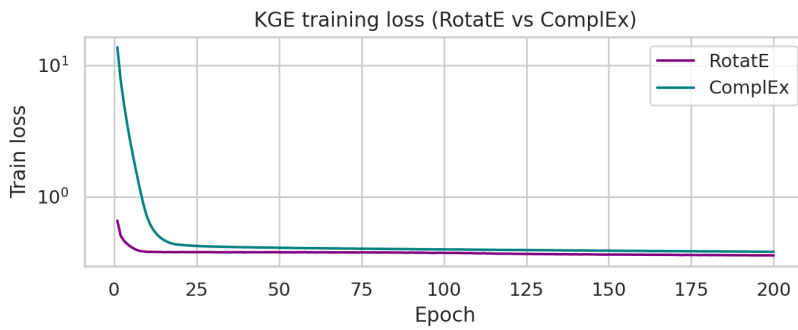


Figure 13. KGE training loss (log scale) for RotatE and ComplEx over 200 epochs. The scales differ between models (ComplEx $\approx$ 13 $\rightarrow$ 3; RotatE $\approx$ 0.7 $\rightarrow$ 0.4); RotatE converges to a lower absolute loss but this does not translate into downstream HGT gains (Table 3 in main text).

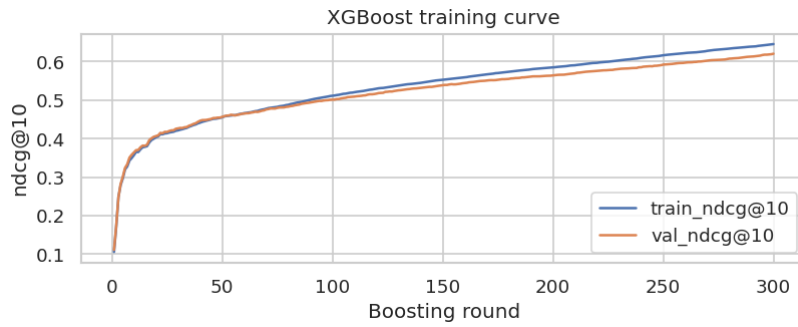


Figure 14. XGBoost LTR training curve (rank: ndcg@10 objective). Early stopping terminates training well before 300 rounds, confirming that the retrieval bottleneck — not under-training — limits ranking performance.

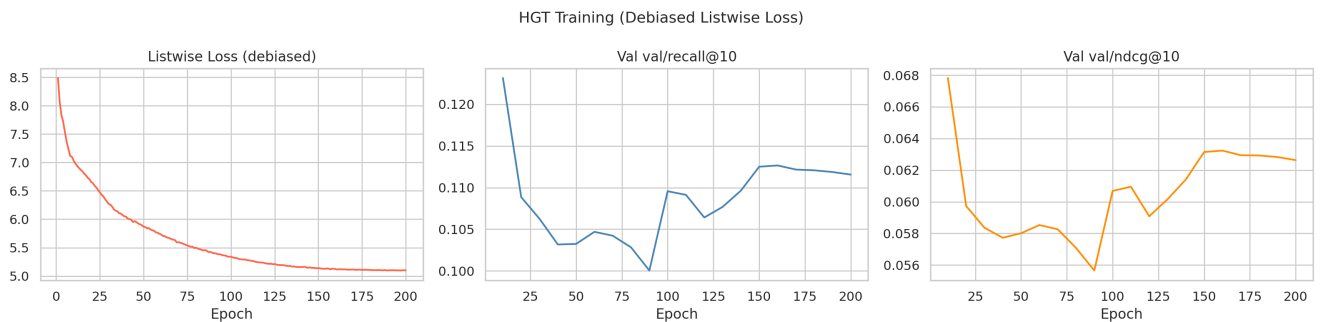


Figure 15. HGT training curves (200 epochs, RotatE init). *Left*: listwise debiased loss decreases monotonically. *Centre*: validation Recall@10. *Right*: validation NDCG@10. All three panels confirm stable optimisation with no divergence.

I. Multi- K Model Comparison

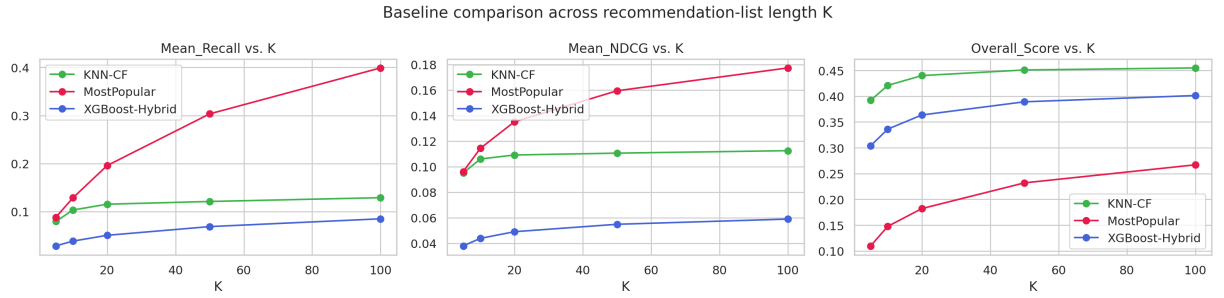


Figure 16. Key metrics at $K \in \{5, 10, 20, 50, 100\}$ for all models. KNN-CF's Overall Score advantage grows with K as its catalogue-coverage advantage compounds. MostPopular leads on Recall and NDCG at every K but achieves near-zero coverage throughout.

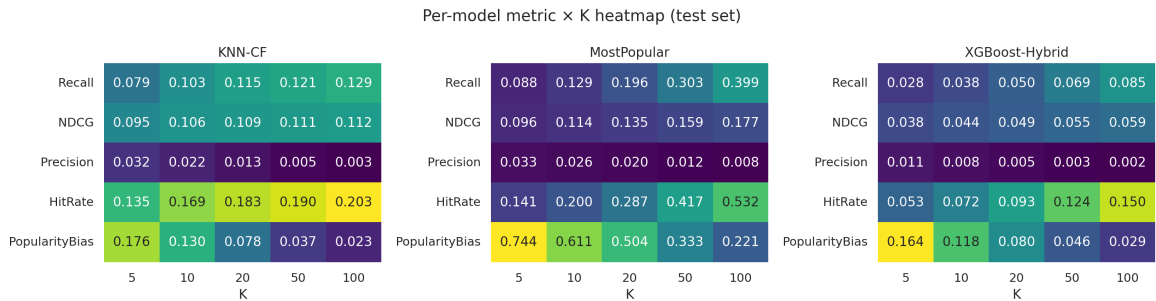


Figure 17. Per-user NDCG@10 and Recall@10 score distributions across all test users. KNN-CF and MostPopular show contrasting shapes: MostPopular concentrates scores at a narrow high peak; KNN-CF distributes mass more broadly at lower per-user NDCG but reaches more users.

J. Population-Level Qualitative Comparison

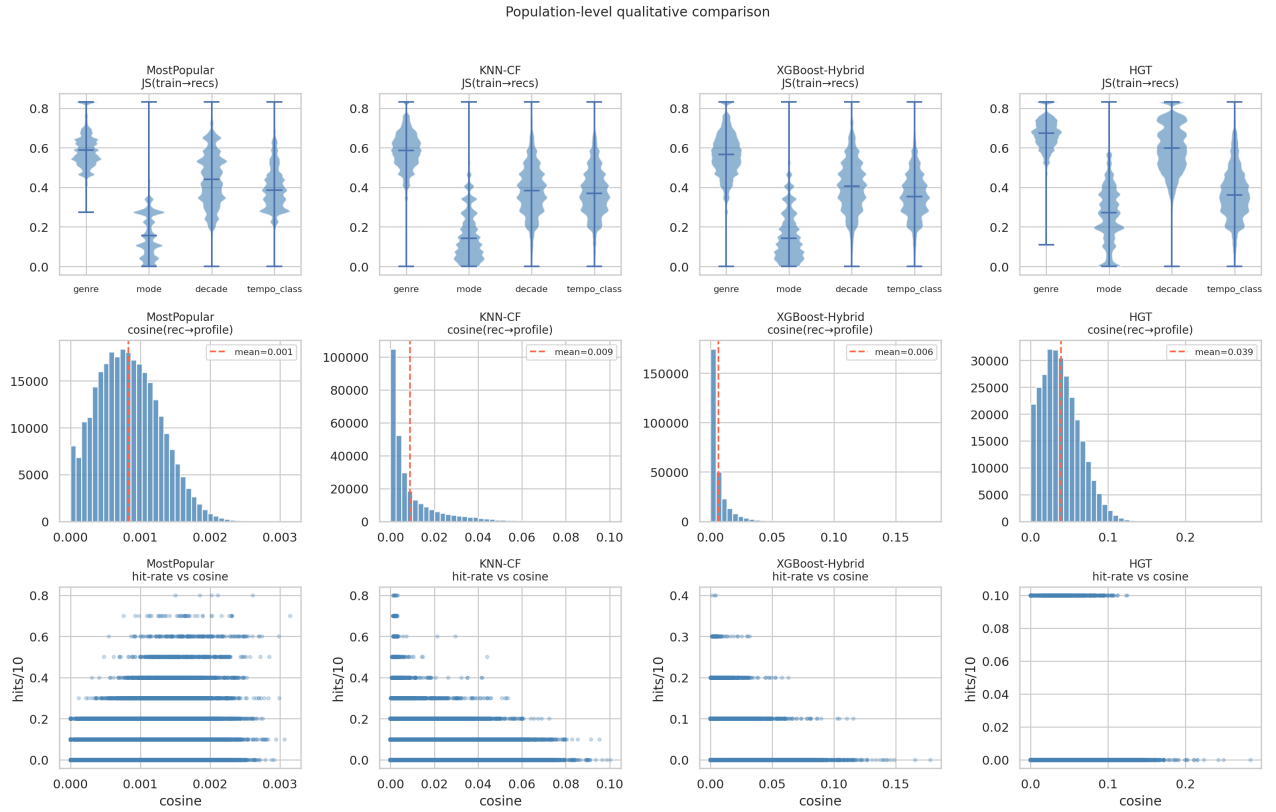


Figure 18. Population-level summary comparing all models (including HGT in self-evaluation mode) across genre diversity, decade coverage, and popularity of recommendations. The HGT produces the most musically diverse recommendations of all evaluated models, consistent with its low Popularity Bias.

K. XGBoost Explainability

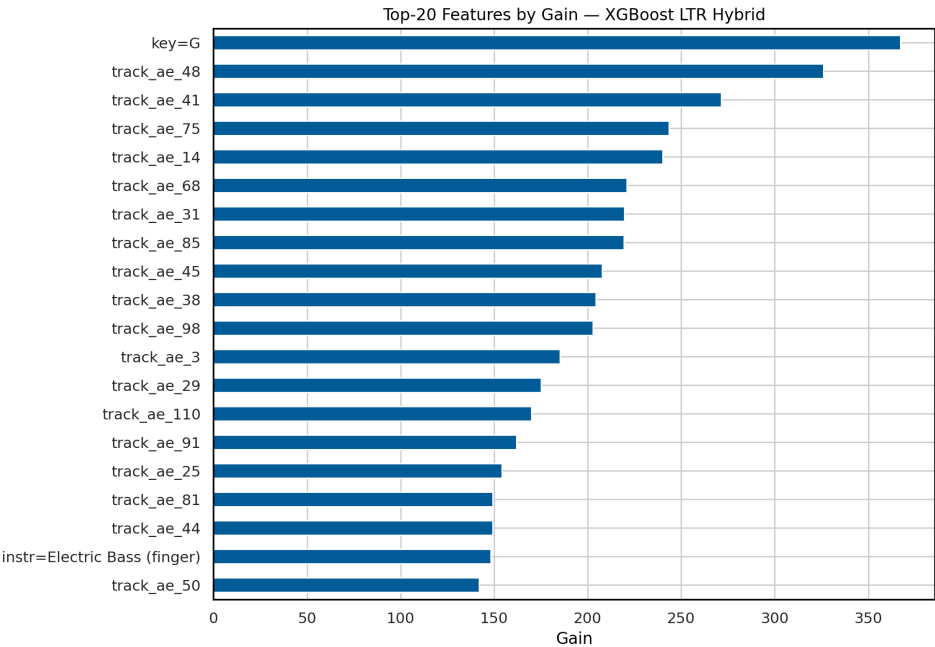


Figure 19. Top-20 XGBoost features by cumulative gain. AE embedding dimensions dominate, with the user-profile AE dimensions contributing more than track-side dimensions, indicating that user preference representation is the primary driver of re-ranking decisions.

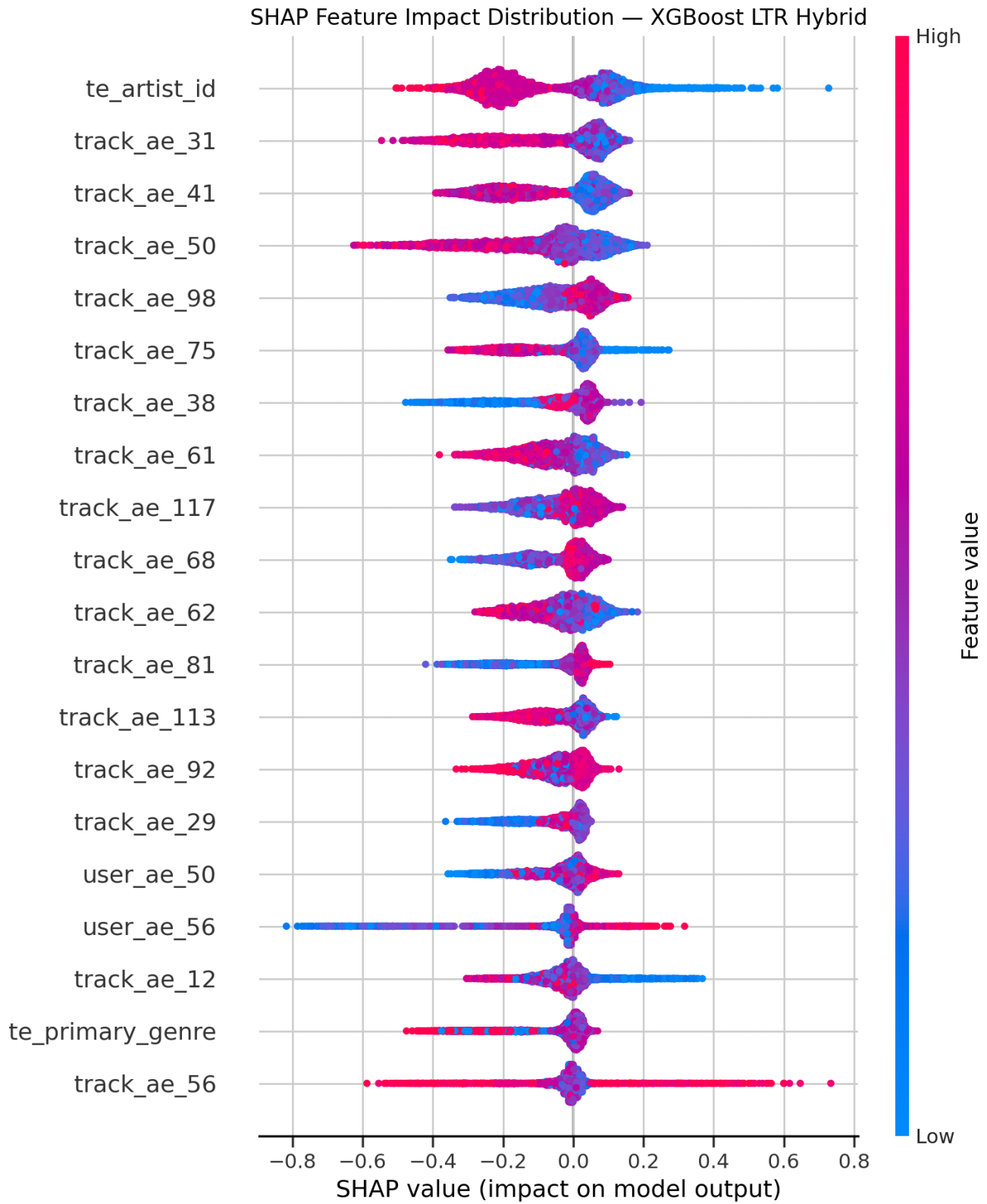


Figure 20. SHAP beeswarm summary over 1,000 sampled test users. Each point is one (user, track) pair; colour encodes feature value. AE embedding dimensions concentrate the largest SHAP magnitudes, confirming that the autoencoder bottleneck is the primary information carrier for re-ranking. Categorical features (genre, key, tempo class) contribute more modestly.

L. HGT Attention Rationales

The following figures show HGT recommendation explanations for three qualitative case studies: a *success-heavy* user (all top recommendations are hits), a *failure* user (top recommendation is a miss), and a *mixed* user. For each case, the first panel traces which of the user's previously listened tracks activated the recommendation via shared attributes; the second panel shows the KG neighbourhood context used by attention.

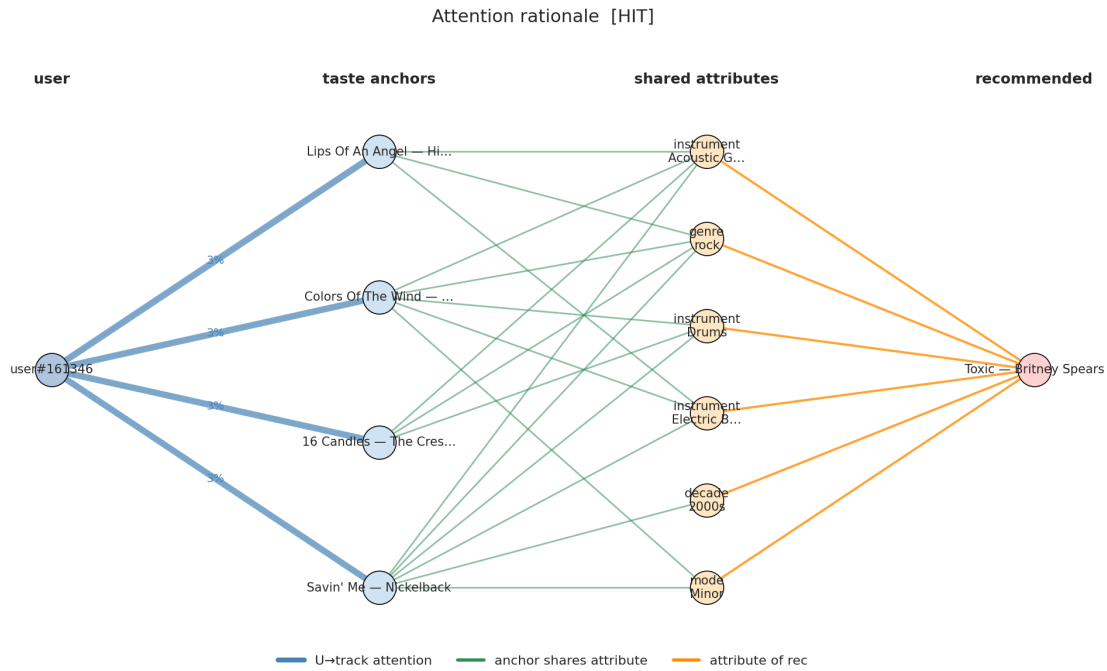


Figure 21. HGT attention rationale — success case: *Toxic* by Britney Spears recommended to user#161346 (HIT ✓, score +0.459). Attention concentrates on tracks sharing the *pop/dance pop* genre and *Allegro* tempo class.

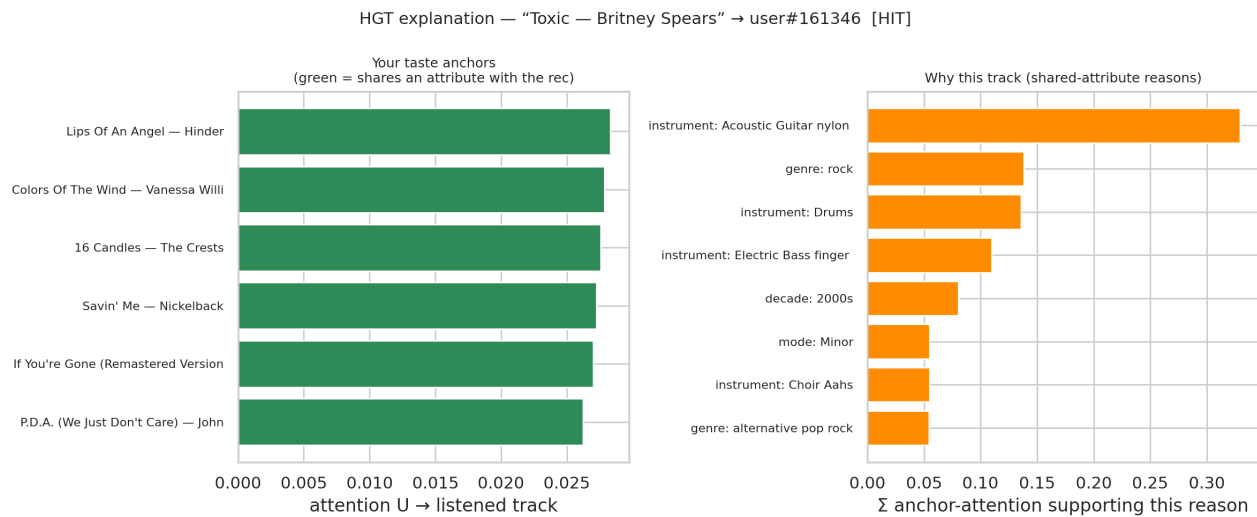


Figure 22. KG neighbourhood context for the success case above, showing the entity types and edge types that contributed most attention mass.

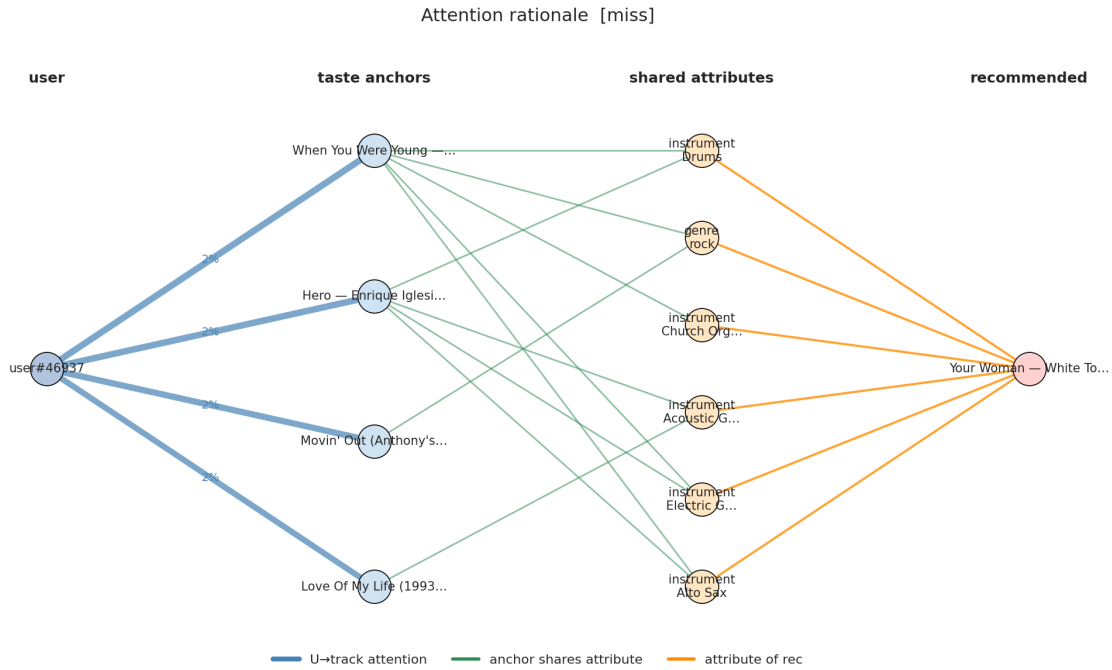


Figure 23. HGT attention rationale — failure case: *Your Woman* by White Town recommended to user#46937 (miss ×, score +0.465). The model latches onto an *Andante/F minor* signature shared with several listened tracks, but the user's preferences do not align with the *indie/alternative hip-hop* genre cluster of the recommended track.

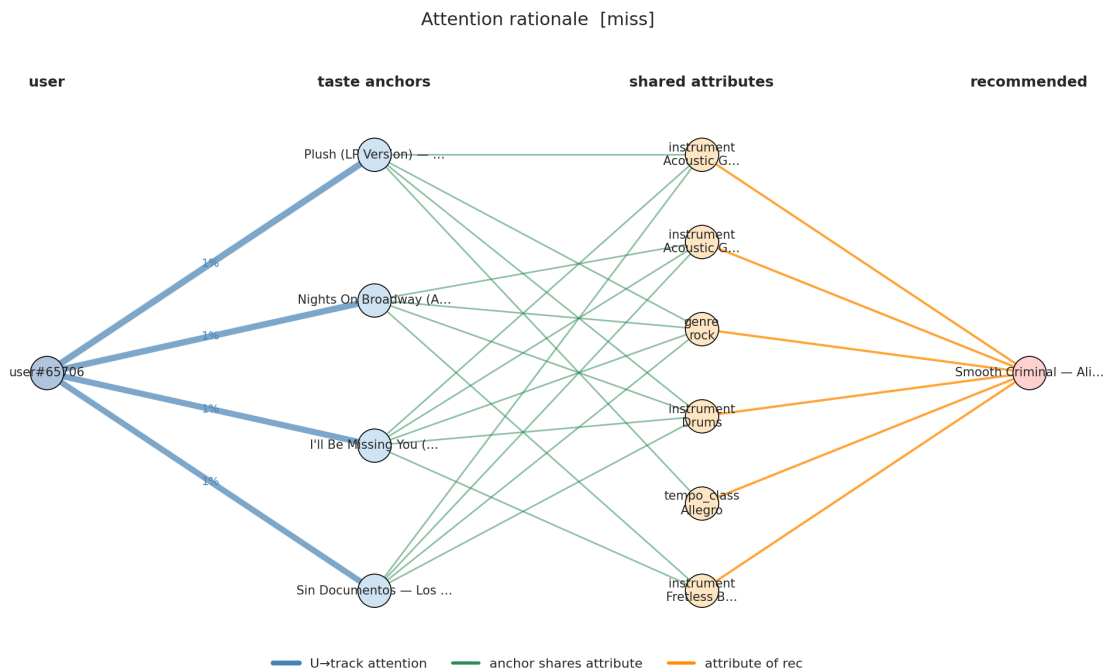


Figure 24. HGT attention rationale — mixed case: *Smooth Criminal* by Alien Ant Farm recommended to user#65706 (miss ×, score +0.359). Attention is spread across multiple edge types, indicating that no single strong signal drove the recommendation.

M. Latent Space Analysis

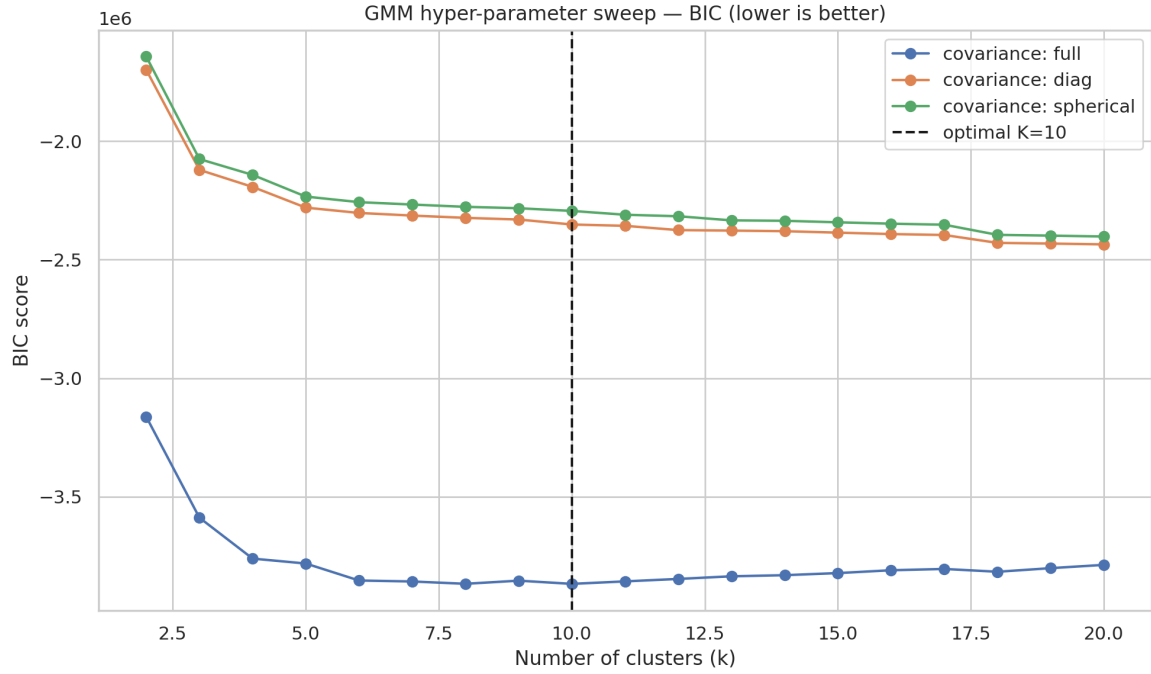


Figure 25. UMAP projection of HGT latent embeddings for all node types. Users (grey), tracks (blue), artists (orange), genres (green), and structural nodes (keys, tempo classes, decades, instruments) occupy distinct but partially overlapping regions, indicating that the HGT learns type-aware representations while preserving semantic proximity.

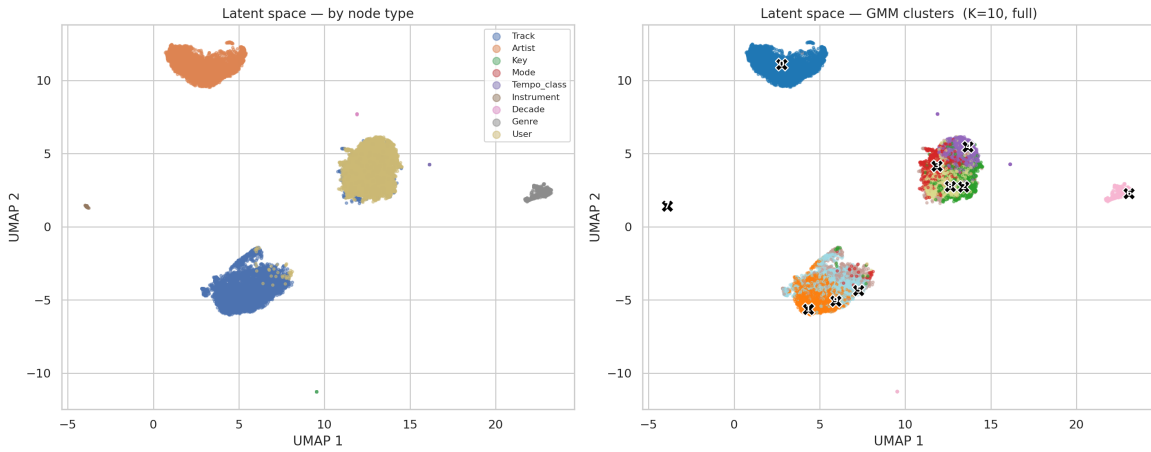


Figure 26. GMM cluster assignments overlaid on the UMAP projection. The BIC-optimal GMM partitions the latent space into interpretable clusters that cross node-type boundaries, suggesting that the HGT has learned shared musical concepts (e.g., genre communities) rather than purely instance-level representations.

N. Qualitative Per-User Model Comparison

Table 8 reports the quantitative profile of five selected users across all models. **cos_mean** and **cos_max** are cosine similarities between the user’s HGT embedding and the recommended tracks’ HGT embeddings (computed uniformly in the HGT latent space for all models). **js_genre** is Jaccard similarity of dominant genre sets between the user’s training profile and the recommended list. **Hits** is the number of test-set ground-truth items appearing in the top-10 recommendation list. The pattern holds across all five cases: HGT achieves substantially higher embedding similarity and genre alignment than any baseline, despite producing fewer or equal hits.

Table 8. Per-user qualitative comparison across all models. cos_mean/cos_max: mean/max cosine similarity (HGT latent space); js_genre: genre Jaccard similarity; Hits: #test-set hits in top-10.

Case	Model	Hits	cos_mean	cos_max	js_genre
success-heavy (u=27917 n_train=47)	MostPopular	4	0.00246	0.00673	0.447
	KNN-CF	4	0.00273	0.00673	0.457
	XGBoost	3	0.00857	0.03933	0.462
	HGT	1	0.03826	0.10988	0.639
failure-1 (u=155906 n_train=81)	MostPopular	0	0.00230	0.00691	0.536
	KNN-CF	0	0.00811	0.03917	0.604
	XGBoost	0	0.00382	0.01888	0.452
	HGT	0	0.07321	0.28695	0.534
failure-2 (u=173477 n_train=67)	MostPopular	0	0.00218	0.00590	0.478
	KNN-CF	0	0.00652	0.01837	0.497
	XGBoost	0	0.01426	0.07578	0.516
	HGT	0	0.09791	0.50601	0.626
mixed-1 (u=86839 n_train=134)	MostPopular	6	0.00271	0.00711	0.509
	KNN-CF	4	0.00635	0.02262	0.498
	XGBoost	1	0.00698	0.01664	0.551
	HGT	0	0.11951	0.70869	0.611
mixed-2 (u=266206 n_train=112)	MostPopular	0	0.00214	0.00582	0.507
	KNN-CF	1	0.00537	0.01281	0.478
	XGBoost	1	0.00296	0.01086	0.491
	HGT	1	0.01403	0.03824	0.665

Note: All cosine similarities are computed in the HGT’s 64-dimensional ℓ_2 -normalised output space. For non-HGT models, this measures how well the recommended items map into the region the HGT associates with the user’s preferences — not a metric those models were optimised for. The consistent advantage of HGT in both embedding similarity and genre alignment, despite lower hit counts, indicates that the model has learned a well-structured preference representation whose quality is not fully captured by offline hit-rate evaluation.